Localization With the Fingerprints of Multipath Angle Delay Spectrum of OFDM Signal

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***Abstract*— Localization is becoming a fundamental requirement for various applications. For many Global Navigation Satellite Systems degraded environments, such as urban and indoor, localization based on the fingerprints of wireless signals has attracted increasing attentions. In this study, a fingerprint, which is based on the multipath channel features of massive multiple input multiple output (MIMO) orthogonal frequency division multiplexing signals, is designed and named as angle delay power spectrum (ADPS). The ADPS is constructed by using the proposed angle domain Fourier transform and delay domain Fourier transform. It can exhibit the signatures of arrival angle and delay of the received signals in the multipath channel more clearly. Next, an adjustable weight weighted K-nearest neighbor (AWKNN) algorithm along with a cosine similarity criterion is utilized to perform the pattern matching localization with the ADPS fingerprints. By introducing the adjustable ability for the K value of the traditional WKNN methods, the AWKNN can obtain a better positioning accuracy. Simulation and real-test results based on public databases prove the superiority of the proposed method to the traditional ones.**

***Index Terms*— Localization with fingerprints, massive MIMO, channel frequency response, ADPS, AWKNN.**

I. Introduction

IN RECENT years, accurate wireless localization has become an essential requirement for many applications. Although Global Navigation Satellite Systems (GNSS) can achieve satisfactory positioning precisions accuracy in many outdoor scenarios, it still suffers from severe performance degradation or fix failure in dense urban environment [1]. Therefore, many wireless signals, such as Wireless Fidelity (WiFi), Global System for Mobile Communications (GSM), Long Term Evolution (LTE), and 5th Generation Mobile Communication Technology (5G), are utilized for localization [2], [3], [4]. These techniques typically adopt the trilateration principle to perform localization with measurements from multiple transmitter nodes [5]. The measurements can be either the time-of-arrival (TOA) [6], the angle-of-arrival (AOA) [7], or the received signal strength (RSS) [8]. However, these localization techniques are normally susceptible to multipath or non-line-of-sight (NLOS) signals, and require precise timing synchronization among transmitter nodes [9].

Another category of wireless localization approaches follows the principle of fingerprint matching. A certain type of signal fingerprint, which consists of parameters associated with position information, is measured at known locations and constructs the fingerprint database. Then, a pattern-matching method based on the fingerprint database can be utilized to estimate the unknown user positions at the phase of online localization.

Traditionally, the RSS was used to form the fingerprint database [10], [11]. However, the inconsistency and non-standardization among different user devices will lead to large positioning errors when taking RSS as fingerprints. Therefore, differential RSS (DRSS) was proposed to obtain a more robust signature in [12] and [13]. Moreover, the combination of DRSS [12] and the hyperbolic location fingerprints [13] is designed to leverage the mutually complementary features of different types of fingerprints [14]. However, the DRSS is prone to fluctuation in mutipath environment, thus leading to inaccurate user position estimates.

Compared with RSS/DRSS fingerprints, multipath channel fingerprints have attracted more attentions because this type of fingerprint contains more rich and accurate information about the wireless signals propagation environment. For example, in [15], [16], [17], [18], and [19] the signal channel state information (CSI) was used as fingerprints because CSI contains both the amplitude and phase information of the channel. Peng et al. [20] proposed a joint fingerprint that is composed of the received signal strength indication (RSSI) and the phase difference of arrival (PDOA). He and So [21] leveraged the time difference of arrival (TDOA) measurement to estimate a coarse location for the user, and then employed the RSS fingerprint to obtain a more accurate position estimate.

As the rapid development of the massive multiple-input multiple-output orthogonal frequency-division multiplexing (MIMO-OFDM) system in recent years, the CSI-based fingerprint localization method for the MIMO-OFDM system attracts much attention and interest by the academia and industry [22]. The large number of the antenna elements in a massive MIMO system can noticeably improve the estimation accuracy for the angle and delay of the OFDM signals. Besides, the channel estimation techniques to acquire CSI for massive MIMO-OFDM system have been studied by many scholars [23], [24], [25], [26]. For examples,

Manuscript received 31 July 2023; revised 19 December 2023 and 11 April 2024; accepted 29 May 2024. Date of publication 11 June 2024; date of current version 11 October 2024. The associate editor coordinating the review of this article and approving it for publication was L. Simic. *(Corresponding author: Xin Chen.)*

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Color versions of one or more figures in this article are available at https://doi.org/10.1109/TWC.2024.3408925.

Digital Object Identifier 10.1109/TWC.2024.3408925

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Pan et al. proposed a compression-based linear minimum mean square error (CLMMSE) channel estimation algorithm for massive MIMO in the 5G system [24]. Jagannatham et al. developed a pilot-aided channel estimation scheme by employing the sparse Bayesian learning framework [25]. Xiang et al. utilized a neural network to estimate the downlink channel based on the uplink cluster data [26]. Therefore, there is strong motivation to implement wireless fingerprint localization based on estimated CSI in massive MIMO-OFDM systems.

In massive MIMO-OFDM systems, Ye et al. [27] built a large-scale CSI as fingerprint for localization. Although the CSI contains the AOA, TOA, and RSS information, these parameters are usually deeply inter-coupled with each other, thus adversely influencing the later pattern matching for localization. Thus, Gao et al. proposed an angle delay channel power matrix (ADCPM) as fingerprint to decouple both the AOA and power delay profile (PDP) information [28], [29], [30]. However, the ADCPM fingerprint is obtained by a fast Fourier transform (FFT) on the averaged channel states over time, which results in low fingerprint resolution. To our best knowledge, fingerprint resolution, which determines the positioning accuracy, has not been proposed in previous studies.

As to the pattern matching for localization, deterministic-model-based algorithms [31], probabilistic-model-based algorithms [32], [33], and machine-learning-based algorithms [29], [34], [35], [36], [37], [38] are all widely adopted. As the fast development of the artificial intelligence technology in recent years, the machine-learning-based method has shown better localization accuracy over the former two types of approaches. These algorithms include K-nearest neighbor (KNN) [34], weight K nearest neighbor (WKNN) [35], convolutional neural network (CNN) [29], [36], artificial neural network (ANN) [37], deep neural network (DNN) [38], etc. Inevitably, the use of neural networks requires much time for training and increases the implementation complexity for the localization method. During these two relatively prolonged procedures, it is probable that the environment changes and the dataset becomes invalid [39]. Compared to small-sized neural network algorithms, the WKNN algorithm performs matching quickly and is robust to the environment.

This paper presents a new type of fingerprint based on the specially designed angle delay power spectrum (ADPS) for a massive MIMO-OFDM signal. The ADPS fingerprint, which is extracted from the channel frequency response (CFR) matrix, can more markedly exhibit the TOA, AOA, and received channel power of an OFDM signal when propagating through a multipath channel. It decouples theses channel parameters embedded in the original CFR, and hence helps improve the localization accuracy. Besides, the theoretical distinguishable resolution boundary for the ADPS fingerprint is derived as well, which can be used as a guidance for the plan of the fingerprint database construction. Furthermore, an adjustable weight WKNN (AWKNN) algorithm is proposed in this study. This algorithm outperforms the traditional WKNN methods by adjusting the key K value in a dynamic way and without the requirement of any prior knowledge of the environment or the fingerprint sampling spacing.

TABLE I
NOTATION DEFINTION

Notation	Defintion
P	Number of path length
N_t	Number of ULA antennas at user
$\mathbf{a}$	ULA response vector at user
ϕ_p	AOA of the pth path
d_s	Antenna element spacing
λ_c	Carrier wavelength
N_c	Number of subcarriers
h_l	CFR of the lth subcarrier
α_p	Complex channel gain of the pth path
$\mathcal{CN}(\cdot)$	Gaussian distribution
σ_p	Gaussian distribution variance of the pth channel gain
n_p	Normalized delay of the pth path
d_p	Distance of the pth path
c	Speed of light
T_s	Sampling interval
$\lfloor\cdot\rceil$	Rounding approximation
$\mathbf{H}$	CFR matrix
$\mathbf{s}$	OFDM symbol sequence in the time domain
$\mathbf{S}$	OFDM symbol sequence in the frequency domain
$\mathbf{Y}$	Received signal
$\mathbf{N_W}$	AWGN noise matrix
$\tilde{\mathbf{N}}_\mathbf{W}$	Estimation errors in the CFR matrix
σ_w^2	Gaussian distribution variance of channel estimation errors
$\mathbf{D}$	ADFT matrix
$\mathbf{F}$	DDFT matrix
ζ_p	Wireless channel gain
K	Number of nearest neighbors in the WKNN algorithm
K_{AWE}	Number of nearest neighbors in the AWKNN algorithm
w_i	Weight factors in the AWKNN algorithm
K_{max}	Maximum value for K_{AWE}
N	Number of grid points
r_{th}	Threshold of two CoADPS

The remainder of this paper is organized as follows: The signal model for a massive MIMO-OFDM system is presented in Section II. The proposed ADPS fingerprint is given in Section III. Next, the theoretical resolution boundary of the ADPS fingerprint is derived in Section IV. The AWKNN algorithm is described in Section V. Performance evaluation is provided in Section VI. Finally, Section VII concludes the paper. In addition, to improve readability, we summarize mathematical symbols used throughout the paper in Table I.

II. SYSTEM MODEL

Assume that an OFDM signal is transmitted from a base station (BS) to a user equipment in a multipath environment as shown in Fig. 1. The number of scatters along the signal propagation path is $P > 1$. The user is equipped with a uniform linear array (ULA) antenna with N_t elements, which has $N_t \gg 1$ for massive MIMO system. The ULA response vector is expressed as $\mathbf{a}(\phi_p) = [a_1(\phi_p), a_2(\phi_p), \ldots, a_{N_t}(\phi_p)]^T$, where $\phi_p \in (0, \pi)$ is the AOA of the pth signal path. The rth element $a_r(\phi_p)$ in $\mathbf{a}(\phi_p)$ is expressed as $e^{-j2\pi\frac{(r-1)d_s\cos(\phi_p)}{\lambda_c}}$, where d_s and λ_c are the antenna element spacing and the carrier wavelength, respectively.

For a MIMO-OFDM system, suppose that the number of subcarriers is N_c. The CFR of the lth $(l = 1, \ldots, N_c)$ subcarrier of the OFDM symbol can be estimated and

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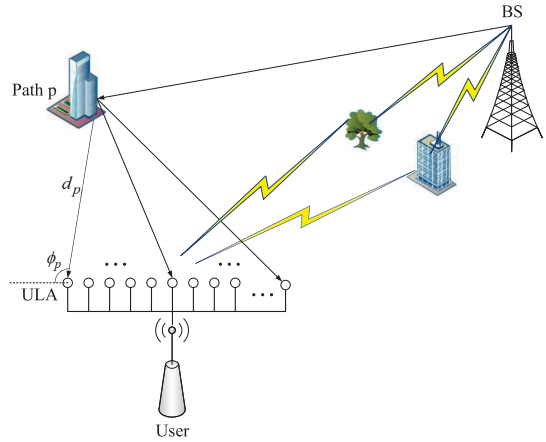

Fig. 1. Illustration for the propagation of a massive MIMO-OFDM signal from the BS to a user in a multipath environment.

expressed as [40]

$$h_l = \sum_{p=1}^{P} \alpha_p \mathbf{a}(\phi_p) e^{-j2\pi \frac{(l-1)n_p}{N_c}}, \tag{1}$$

where α_p denotes the complex channel gain for the pth path, which follows a Gaussian distribution $\mathcal{CN}(0, \sigma_p)$ with a zero mean and a variance of σ_p. $n_p = \lfloor \frac{d_p}{cT_s} \rceil$ is the normalized temporally resolvable propagation delay of the pth signal path, where d_p is the pth path distance from the BS to the first received antenna element of the ULA antenna, c and T_s are the speed of light and the sampling interval, respectively. $\lfloor \cdot \rceil$ denotes the rounding approximation. Without loss of generality, the multipath scatters are assumed to be uniformly and independently distributed in the coverage area. Thus, the AOAs ϕ_p and the delays n_p of different paths can be taken as independent with each other.

The matrix form of the CFR of all subcarriers can be written as

$$\mathbf{H} = \begin{bmatrix} h_{11} & \cdots & h_{1,N_c} \\ \vdots & \ddots & \vdots \\ h_{N_t,1} & \cdots & h_{N_t,N_c} \end{bmatrix}, \tag{2}$$

where $h_{r,l} = \sum_{p=1}^{P} \alpha_p e^{-j2\pi \frac{d_s(r-1)\cos\phi_p}{\lambda_c}} e^{-j2\pi \frac{(l-1)n_p}{N_c}}, r = 1, \ldots, N_t, l = 1, \ldots, N_c$. This matrix $\mathbf{H}$ spans N_t rows and N_c columns, with N_t being the number of ULA antennas and N_c the number of subcarriers.

Denote $\mathbf{s} = [s(0), s(1), \ldots, s(N_c-1)]^T$ in an OFDM block transmitted in the time domain. Through a DFT transform operation, the transmitted sequence can be expressed in the frequency domain as $\mathbf{S} = [S(0), S(1), \ldots, S(N_c-1)]^T$. Then the received signal by the user can be expressed as

$$\mathbf{Y} = \mathbf{HS} + \mathbf{N_W} \tag{3}$$

where $\mathbf{N_W}$ response the AWGN noise matrix.

Many methods have been proposed to estimate the CFR matrix of a multipath channel for the MIMO-OFDM system [23], [24], [25], [26]. In the conventional pilot channel estimation method, the frequency domain channel estimation for the pilot position based on least squares criterion is given by

$$\tilde{\mathbf{H}} = \mathbf{YS^{-1}} = \mathbf{H} + \tilde{\mathbf{N}}_\mathbf{W} \tag{4}$$

where $\tilde{\mathbf{N}}_\mathbf{W} = \mathbf{N_W S^{-1}}$ represents the estimation errors in the CFR matrix $\mathbf{H}$. The elements in $\tilde{\mathbf{N}}_\mathbf{W}$ are distributed as $\mathcal{CN}(0, \sigma_w^2)$ with a zero mean and a variance of σ_w^2. Especially, for a massive MIMO-OFDM system, it has been shown that the channel estimation error can be well mitigated due to the usage of the large number of antenna elements [41]. In the following derivation, we simply assume that the CFR has already been obtained through a proper channel estimation method.

III. Design of The ADPS Fingerprint

According to (1) and (2), it can be seen that the CFR contains the channel information about the AOAs, delays, and channel gains of the multiple reflected paths of an OFDM signal. The channel information is further related to the positions at which the user is located. Therefore, the CFR matrix is reasonably able to be taken as a signal fingerprint for user positioning. However, it may not be the best choice to use the original CFR matrix as the fingerprint because the AOAs, delays, and channel gains of a signal path are deeply inter-coupled with each other, as a result leading to obscure the relationship of the characteristics of the CFR with the locations. Therefore, in this study a new fingerprint format, which is derived from the original CFR matrix, is proposed and named as ADPS. The ADPS fingerprint is composed of two submatrices, which are the angle power spectrum (APS) matrix and the delay power spectrum (DPS) matrix, respectively. The APS matrix is obtained by the proposed angle domain Fourier transform (ADFT) on CFR, and the DPS matrix is obtained by the proposed delay domain Fourier transform (DDFT) on CFR. The final ADPS can markedly exhibit the angle and delay features of a multipath channel and reveal the relationship of CSI with locations. The detailed derivation of ADPS is given in the following subsections.

A. The Derivation of APS Matrix

The APS matrix is obtained by the multiplication of the CFR matrix, given in (2), with the ADFT matrix, which is designed as

$$\mathbf{D} = \begin{bmatrix} d_{11} & \cdots & d_{1,N_t} \\ \vdots & \ddots & \vdots \\ d_{N_t,1} & \cdots & d_{N_t,N_t} \end{bmatrix}, \tag{5}$$

where

$$[\mathbf{D}_{N_t \times N_t}]_{m,r} = \frac{1}{\sqrt{N_t}} e^{-j2\pi \frac{d_s(r-1)\cos\pi\frac{m}{N_t}}{\lambda_c}}. \tag{6}$$

Here $[\cdot]_{m,r}$ denotes the mth row and rth column element of the matrix $[\cdot]$, and $m, r = 1, \ldots, N_t$. Then, the APS

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is given by

$$\begin{aligned}\mathrm{APS} &= \mathbf{DH}\\ &= \begin{bmatrix} d_{11} & \cdots & d_{1,N_t}\\ \vdots & \ddots & \vdots\\ d_{N_t,1} & \cdots & d_{N_t,N_t}\end{bmatrix}\begin{bmatrix} h_{11} & \cdots & h_{1,N_c}\\ \vdots & \ddots & \vdots\\ h_{N_t,1} & \cdots & h_{N_t,N_c}\end{bmatrix}\\ &= \begin{bmatrix} \sum_{r=1}^{N_t} d_{1,r}h_{r,1} & \cdots & \sum_{r=1}^{N_t} d_{1,r}h_{r,N_c}\\ \vdots & \ddots & \vdots\\ \sum_{r=1}^{N_t} d_{N_t,r}h_{r,1} & \cdots & \sum_{r=1}^{N_t} d_{N_t,r}h_{r,N_c}\end{bmatrix}\\ &= [\sum_{r=1}^{N_t} d_{m,r}h_{r,l}]_{N_t\times N_c}, \end{aligned} \tag{7}$$

where $[\cdot]_{N_t\times N_c}$ denotes a matrix with a size of $N_t \times N_c$.

Then, the element in the APS matrix is derived as follows

$$\begin{aligned}&\sum_{r=1}^{N_t} d_{m,r}h_{r,l}\\ &= \sum_{p=1}^{P}\alpha_p\frac{1}{\sqrt{N_t}}e^{-j2\pi\frac{ln_p}{N_c}}\sum_{r=0}^{N_t-1}e^{-j2\pi r\frac{d_s(\cos\phi_p+\cos\pi\frac{m}{N_t})}{\lambda_c}}\\ &= \sum_{p=1}^{P}\alpha_p\frac{1}{\sqrt{N_t}}e^{-j2\pi\frac{ln_p}{N_c}}\left(\frac{1-e^{-j2\pi N_t\frac{d_s(\cos\phi_p+\cos\pi\frac{m}{N_t})}{\lambda_c}}}{1-e^{-j2\pi\frac{d_s(\cos\phi_p+\cos\pi\frac{m}{N_t})}{\lambda_c}}}\right)\\ &= \sum_{p=1}^{P}\alpha_p\frac{1}{\sqrt{N_t}}e^{-j2\pi\frac{ln_p}{N_c}}e^{-j(N_t-1)\pi\frac{d_s(\cos\phi_p+\cos\pi\frac{m}{N_t})}{\lambda_c}}\\ &\quad\times W_{N_t}\left(\pi\frac{d_s(\cos\phi_p+\cos\pi\frac{m}{N_t})}{\lambda_c}\right), \end{aligned} \tag{8}$$

where $W_{N_t}(\omega) = \frac{\sin(N_t\omega)}{\sin(\omega)}$, $m, r = 0,\ldots,N_t-1$, $l = 0,\ldots,N_c-1$.

Theorem 1: It can be proved that if N_t is a significantly large number, $\sum_{r=1}^{N_t} d_{m,r}h_{r,l}$ can be approximated as

$$\sum_{r=1}^{N_t} d_{m,r}h_{r,l} = \sum_{p=1}^{P}\alpha_p\sqrt{N_t}e^{-j2\pi\frac{ln_p}{N_c}}\delta(m-N_t(1-\frac{\phi_p}{\pi})). \tag{9}$$

Proof: The detailed derivations for Theorem 1 are given in Appendix A. □

Substitute (9) into (7), the APS can be written as

$$\mathrm{APS} = [\sum_{\mathrm{p}=1}^{\mathrm{P}}\alpha_{\mathrm{p}}\sqrt{\mathrm{N_t}}\mathrm{e}^{-\mathrm{j}2\pi\frac{\mathrm{ln_p}}{\mathrm{N_c}}}\delta(\mathrm{m}-\mathrm{N_t}(1-\frac{\phi_{\mathrm{p}}}{\pi}))]_{\mathrm{N_t}\times\mathrm{N_c}}. \tag{10}$$

It is seen that the peaks in the APS matrix occur at the arrival angles of the major signal paths, including both the line of sight (LOS) and multipath.

B. The Derivation of DPS Matrix

The DPS matrix is obtained by the multiplication of the CFR matrix with the DDFT matrix, which is designed as

$$\mathbf{F} = \begin{bmatrix} f_{11} & \cdots & f_{1,N_c}\\ \vdots & \ddots & \vdots\\ f_{N_c,1} & \cdots & f_{N_c,N_c}\end{bmatrix}, \tag{11}$$

where

$$[\mathbf{F}_{N_c\times N_c}]_{l,n} = \frac{1}{\sqrt{N_c}}e^{-j2\pi\frac{(l-1)n}{N_c}}. \tag{12}$$

Here $l, n = 1,\ldots,N_c$. The DPS is given as follows

$$\begin{aligned}\mathrm{DPS} &= \mathbf{HF}\\ &= \begin{bmatrix} h_{11} & \cdots & h_{1,N_c}\\ \vdots & \ddots & \vdots\\ h_{N_t,1} & \cdots & h_{N_t,N_c}\end{bmatrix}\begin{bmatrix} f_{11} & \cdots & f_{1,N_c}\\ \vdots & \ddots & \vdots\\ f_{N_c,1} & \cdots & f_{N_c,N_c}\end{bmatrix}\\ &= \begin{bmatrix} \sum_{l=1}^{N_c} h_{1,l}f_{l,1} & \cdots & \sum_{l=1}^{N_c} h_{1,l}f_{l,N_c}\\ \vdots & \ddots & \vdots\\ \sum_{l=1}^{N_c} h_{N_t,l}f_{l,1} & \cdots & \sum_{l=1}^{N_c} h_{N_t,l}f_{l,N_c}\end{bmatrix}\\ &= [\sum_{l=1}^{N_c} h_{r,l}f_{l,n}]_{N_t\times N_c}. \end{aligned} \tag{13}$$

Here

$$\begin{aligned}&\sum_{l=1}^{N_c} h_{r,l}f_{l,n}\\ &= \sum_{p=1}^{P}\frac{1}{\sqrt{N_c}}a_pe^{-j2\pi\frac{d_sr\cos\phi_p}{\lambda_c}}\sum_{l=0}^{N_c-1}e^{-j2\pi l(\frac{n_p}{N_c}+\frac{n}{N_c})}\\ &= \sum_{p=1}^{P}\frac{1}{\sqrt{N_c}}\alpha_pe^{-j2\pi\frac{d_s\omega\cos\phi_p}{\lambda_c}}e^{-j(N_c-1)\pi(\frac{n_p}{N_c}+\frac{n}{N_c})}\\ &\quad\times W_{N_c}\left(\pi(\frac{n_p}{N_c}+\frac{n}{N_c})\right), \end{aligned} \tag{14}$$

where $W_{N_c}(d) = \frac{\sin(N_cd)}{\sin(d)}$ and $l, n = 0,\ldots,N_c-1$.

Theorem 2: The aforementioned equation can be simplified as

$$\sum_{l=1}^{N_c} h_{r,l}f_{l,n} = \sum_{p=1}^{P}\sqrt{N_c}\alpha_pe^{-j2\pi\frac{d_sr\cos\phi_p}{\lambda_c}}\delta(n-(N_c-n_p)). \tag{15}$$

Proof: The proof is omitted due to space limitation. The derivation method similar to that used by [Theorem 1, (9)]. □

Then, the DPS can be written as

$$\mathrm{DPS} = [\sum_{p=1}^{P}\sqrt{N_c}\alpha_pe^{-j2\pi\frac{d_sr\cos\phi_p}{\lambda_c}}\delta(n-(N_c-n_p))]_{N_t\times N_c}. \tag{16}$$

The peaks of the DPS happen at the delays of the major signal paths corresponding to the LOS and NLOS scenes in the urban and indoor.

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C. The Derivation of ADPS Matrix

According to the derived APS and DPS, the form of the ADPS fingerprint is developed as

$$\begin{aligned}&\text{ADPS}\\&=\text{APS}\odot\text{DPS}=\mathbf{DH}\odot\mathbf{HF}\\&=\left[\sum_{p=1}^{P}\zeta_p\delta(m-N_t(1-\frac{\phi_p}{\pi}))\delta(n-(N_c-n_p))\right]_{N_t\times N_c},\end{aligned}\tag{17}$$

where $\zeta_p=\sum_{p=1}^{P}\sqrt{N_tN_c}\alpha_p^2e^{-j2\pi\frac{ln_p}{N_c}}e^{-j2\pi\frac{d_sr\cos\phi_p}{\lambda_c}}$.

In (17), ζ_p denotes the wireless channel gain at the specific positions. According to [42], the quantity $|\zeta_p|$ is named as the instantaneous multipath power delay profile of a channel. It can be seen that the proposed ADPS is able to reveal the AOA, TOA, and channel gain information more clearly than the original CFR matrix. When the ADPS is used as the signal fingerprint, it can be expected that more accurate localizations will be obtained.

Remark 1: The core idea of the proposed ADPS fingerprint is to extract AOA and TOA information by using the proposed 'DFT matrices'. This approach can be effectively extended to URA antenna arrays.

In two-dimensional positioning, ULA antennas provide higher resolution in angular detection, resulting in better positioning performance. Uniform rectangular array (URA) antennas offer three-dimensional angle resolution, allowing for differentiation in the horizontal, vertical, and height dimensions. Therefore, URA antennas are a more suitable choice for three-dimensional positioning. Additionally, URA antenna arrays come with higher construction costs, in height-insensitive positioning systems, deploying ULA antennas is a more energy-efficient and superior choice.

IV. The Resolution of the ADPS Fingerprints

The analysis of the spatial differentiable resolution for the proposed ADPS fingerprint is of help to predict the possible accuracy bound. According to (17), the ADPS matrix is composed of two sub-matrices, i.e., the APS and the DPS, respectively. By definition, the APS matrix determines the angular differentiable resolution for two incident paths from close arrival directions, and the DPS matrix determines the delay differentiable resolution for two paths arriving at similar time instants.

With the limited N_t, the APS matrix can be defined as

$$\begin{aligned}\text{APS}=[\sum_{p=1}^{P}\alpha_p\frac{1}{\sqrt{N_t}}e^{-j2\pi\frac{ln_p}{N_c}}e^{-j(N_t-1)\pi\frac{d_s(\cos\phi_p+\cos\pi\frac{m}{N_t})}{\lambda_c}}\\\times W_{N_t}\left(\pi\frac{d_s(\cos\phi_p+\cos\pi\frac{m}{N_t})}{\lambda_c}\right)]_{N_t\times N_c}.\end{aligned}\tag{18}$$

Due to the priori information about fingerprint positioning and collection devices, N_t, N_c, d_s and λ_c are the determined values. Furthermore, when considering the distinguishability of different AOAs in APS, the influence of coefficients $\alpha_p\frac{1}{\sqrt{N_t}}e^{-j2\pi\frac{ln_p}{N_c}}$ can be ignored, and the above formula for APS is considered as a function of variable m. Without loss of generality, we assume there are only two incident paths in the received signal. Thus, the $\text{APS}_R(m)$ can be defined as

$$\begin{aligned}&\text{APS}_R(m)\\&=e^{-j(N_t-1)\frac{\pi d_s(\cos\phi_{p_i}+\cos\frac{\pi m}{N_t})}{\lambda_c}}W_{N_t}(\frac{\pi d_s(\cos\phi_{p_i}+\cos\frac{\pi m}{N_t})}{\lambda_c})\\&\quad+e^{-j(N_t-1)\frac{\pi d_s(\cos\phi_{p_j}+\cos\frac{\pi m}{N_t})}{\lambda_c}}W_{N_t}(\frac{\pi d_s(\cos\phi_{p_j}+\cos\frac{\pi m}{N_t})}{\lambda_c}),\end{aligned}\tag{19}$$

where ϕ_{p_i} and ϕ_{p_j} represent the AOAs of two arbitrarily different wireless transmission paths, and $m=0,\ldots,N_t-1$.

Additionally, due to the complex function property of $\text{APS}_R(m)$, both the envelope superposition effect and the influence of the phase frequency property of $\text{APS}_R(m)$ should be considered to obtain the angular differentiable resolution. The square spectrum of $\text{APS}_R(m)$ can be expressed as

$$\begin{aligned}|\text{APS}_R(m)|^2&=\text{APS}_R(m)\text{APS}_R(m)^H\\&=W_{N_t}^2\left(\frac{\pi d_s(\cos\phi_{p_i}+\cos\frac{\pi m}{N_t})}{\lambda_c}\right)\\&\quad+W_{N_t}^2\left(\pi\frac{d_s(\cos\phi_{p_j}+\cos\frac{\pi m}{N_t})}{\lambda_c}\right)\\&\quad+2\cos\left(\frac{\pi d_s(N_t-1)(\cos\phi_{p_i}-\cos\phi_{p_j})}{\lambda_c}\right)\\&\quad\times W_{N_t}\left(\frac{\pi d_s(\cos\phi_{p_i}+\cos\pi\frac{m}{N_t})}{\lambda_c}\right)W_{N_t}\\&\quad\times\left(\frac{\pi d_s(\cos\phi_{p_j}+\cos\frac{\pi m}{N_t})}{\lambda_c}\right).\end{aligned}\tag{20}$$

It can be seen from (20) that the $|\text{APS}_R(m)|^2$ is composed of two main terms and a cross term. To distinguish two different AOAs ϕ_{p_i} and ϕ_{p_j}, there should be two distinct spectral peaks that appear for function $|\text{APS}_R(m)|^2$ as m varies. Therefore, on the one hand, the two main lobes of two main terms for $|\text{APS}_R(m)|^2$ do not overlap. On the other hand, when two primary terms overlap, the value at the point of intersection between the two terms is lower than the highest point of either term alone.

Here, we define the angular resolution as the condition when APS is able to differentiate between two distinct AOAs.

Theorem 3: The angle resolution of APS *is as follows*

$$\begin{cases}\left|\cos\phi_{p_i}-\cos\phi_{p_j}\right|\geq\dfrac{2}{N_t\frac{d_s}{\lambda_c}}.\\\dfrac{2}{3(N_t-1)\frac{d_s}{\lambda_c}}<\left|\cos\phi_{p_i}-\cos\phi_{p_j}\right|<\dfrac{4}{3(N_t-1)\frac{d_s}{\lambda_c}}.\end{cases}\tag{21}$$

Proof: The detailed derivations for Theorem 3 are given in Appendix B. □

It should be noted that the $\text{APS}_R(m)$ only has values on integer points, thus the actual resolution obtained will be lower than the above results.

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Next, the delay resolution of DPS is analyzed. With limited N_c, the DPS matrix is given as follows

$$\mathrm{DPS}_R(n) = \sum_{p=1}^{P} \frac{1}{\sqrt{N_c}} \alpha_p e^{-j2\pi \frac{d_s r \cos\phi_p}{\lambda_c}} e^{-j(N_c-1)\pi(\frac{n_p}{N_c}+\frac{n}{N_c})} \times W_{N_c}\left(\pi(\frac{n_p}{N_c}+\frac{n}{N_c})\right). \quad (22)$$

For simplicity but without loss of generality, we mainly consider the resolution of two wireless transmission paths. Thus, the $\mathrm{DPS}_R(n)$ can be rewritten as follows

$$\begin{aligned}&\mathrm{DPS}_R(n)\\ &= e^{-j(N_c-1)\pi(\frac{n_{p_i}}{N_c}+\frac{n}{N_c})} W_{N_c}\left(\pi(\frac{n_{p_i}}{N_c}+\frac{n}{N_c})\right)\\ &\quad + e^{-j(N_c-1)\pi(\frac{n_{p_j}}{N_c}+\frac{n}{N_c})} W_{N_c}\left(\pi(\frac{n_{p_j}}{N_c}+\frac{n}{N_c})\right),\end{aligned} \quad (23)$$

where n_{p_i} and n_{p_j} represent the TOAs of two arbitrarily different wireless transmission paths.

Thus, the square spectrum of $\mathrm{DPS}_R(n)$ can be expressed as

$$\begin{aligned}&|\mathrm{DPS}_R(n)|^2 = \mathrm{DPS}_R(n)\mathrm{DPS}_R(n)^H\\ &= W_{N_c}^2\left(\pi(\frac{n_{p_i}}{N_c}+\frac{n}{N_c})\right) + W_{N_c}^2\left(\pi(\frac{n_{p_j}}{N_c}+\frac{n}{N_c})\right)\\ &\quad + 2\cos\left(\pi(N_c-1)(\frac{n_{p_i}}{N_c}-\frac{n_{p_j}}{N_c})\right)\\ &\quad \times W_{N_c}\left(\pi(\frac{n_{p_i}}{N_c}+\frac{n}{N_c})\right) W_{N_c}\left(\pi(\frac{n_{p_j}}{N_c}+\frac{n}{N_c})\right).\end{aligned} \quad (24)$$

To distinguish two different delays n_{p_i} and n_{p_j}, there should be two distinct spectral peaks that appear for function $|\mathrm{DPS}_R(n)|^2$ as n varies. Here, the condition when DPS can distinguish two distinct delays is defined as delay resolution.

Theorem 4: The delay resolution is as follows

$$\begin{cases} |n_{p_i} - n_{p_j}| \geq 2. \\ \dfrac{2N_c}{3(N_c-1)} < |n_{p_i} - n_{p_j}| < \dfrac{4N_c}{3(N_c-1)}. \end{cases} \quad (25)$$

Proof: The derivation process is similar to angular resolution. A similar proof can be found in Appendix B. □

It should be noted that the $\mathrm{DPS}_R(n)$ only has values on integer points, thus the actual resolution obtained will be lower than the above results. From the analysis results of the resolution, it can be seen that the selection of the number of array antenna elements is of great significance to improve the positioning accuracy.

Fig. 2 shows the ADPS matrix with normalized delays $n_p = \{20, 40\}$ and AOAs $\phi_p = \{45°, 90°\}$. The number of ULA antenna elements is $N_t = 60$ with $\frac{d_s}{\lambda_c} = 0.5$ and the number of subcarriers is $N_c = 60$. As shown in (17), the ADPS matrix has the peak points $\sum_{p=1}^{P} \zeta_p$ when $n = N_c - n_p$ and $m = N_t(1 - \frac{\phi_p}{\pi})$. Thus, the row, column and amplitude of ADPS are correspond to TOAs, AOAs and normalized channel power, respectively. It is obvious that n_p and ϕ_p satisfy (21) and (25) with $|n_{p_1} - n_{p_2}| = 20 \geq 2$ and

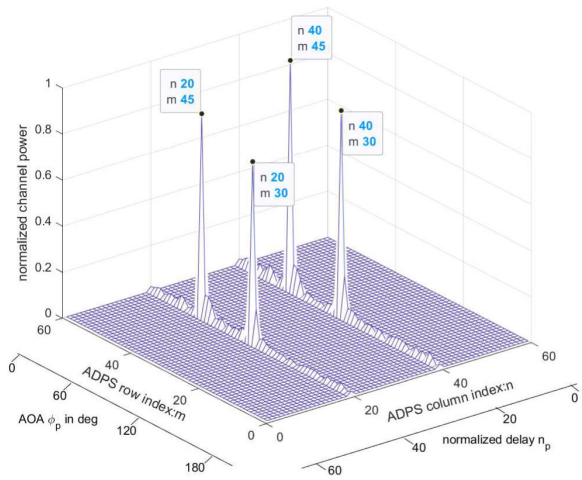

Fig. 2. An illustration of the ADPS fingerprint with with $n_p = \{20, 40\}$ and $\phi_p = \{45°, 90°\}$.

$|\cos\phi_{p_1} - \cos\phi_{p_2}| = 0.7453 \geq \frac{2}{N_t \frac{d_s}{\lambda_c}} = 0.0625$. Both AOAs and normalized delay are distinguishable. The peak points occurs when $m = 30, 45$ and $n = 20, 40$.

V. AWKNN Localization Algorithm Based on ADPS Fingerprints

In this section, an AWKNN algorithm is presented to realize the localization operation based on ADPS fingerprints.

A. The Cosine Similarity Between ADPS Fingerprints

Typically, localization based on fingerprints is a pattern recognition task, which needs similarity evaluations between different sets of fingerprints in order to estimate the most possible position of the user. In the traditional fingerprint location algorithm, the Euclidean distance is usually adopted to measure the difference between two fingerprints [43]. However, cosine similarity [44] is preferred over Euclidean distance as a similarity metric for several reasons. Euclidean distance is sensitive to scaling, while cosine similarity is independent of the length of the fingerprint vectors. Furthermore, the CFR is usually a sparse matrix, and cosine similarity is more suitable for sparse data sets, as it is insensitive to zero values and provides a more accurate measurement of similarity.

For this reason, the cosine similarity evaluation coefficient of ADPS (CoADPS), is utilized to characterize the degree of similarity between two ADPS fingerprints

$$\begin{aligned}\mathrm{CoADPS} &= \cos(\angle(\mathrm{ADPS}_i, \mathrm{ADPS}_j))\\ &= \frac{\sum_{m=1}^{N_t}\sum_{n=1}^{N_c} (\mathrm{ADPS}_i)_{m,n}(\mathrm{ADPS}_j)_{m,n}}{\sqrt{\sum_{m=1}^{N_t}\sum_{n=1}^{N_c} (\mathrm{ADPS}_i)_{m,n}^2}\sqrt{\sum_{m=1}^{N_t}\sum_{n=1}^{N_c} (\mathrm{ADPS}_j)_{m,n}^2}}\\ &= \frac{\sum_{p_i}^{P_i}\sum_{p_j}^{P_j} \zeta_{p_i}\zeta_{p_j}\delta(\phi_{p_i}-\phi_{p_j})\delta(\phi_{p_i}-\phi_{p_j})}{\sum_{p_j}^{P_j}\sum_{p_i}^{P_i} \zeta_{p_i}\zeta_{p_j}}.\end{aligned} \quad (26)$$

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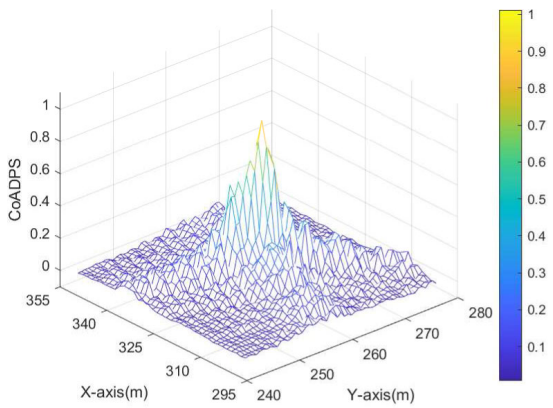

Fig. 3. Example of CoADPS distribution.

where ADPS_i and ADPS_j denote ADPS fingerprint matrix at locations i and j, separately, and $(\mathrm{ADPS}_i)_{m,n}$ and $(\mathrm{ADPS}_j)_{m,n}$ denote the mth row's and nth column's elements of ADPS_i and ADPS_j matrix, respectively. The CoADPS value is a decreasing function about the geometrical distance between locations i and j. Therefore, it can be used as a good metric for indicating the real distances of the unknown estimated location with the reference locations.

In Fig. 3, the trend of the CoADPS variations between a certain position's ADPS fingerprint and nearby positions' fingerprints is illustrated. In this test, the Deep MIMO database [45] is used, which is a simulated communication channel database for either outdoor urban or indoor environment. In this simulation, an outdoor urban signal with 3.5 GHz carrier frequency and 100 MHz bandwidth is assumed to get the channel frequency responses. The X/Y axis stands for the coordinates of the positions in a local coordinate system, and the Z axis shows the values of CoADPS. It can be seen that the CoADPS follows a decreasing trend as the distance of evaluation position is far from the reference position.

B. The AWKNN Localization Algorithm

Fingerprint-based localization can be summarized as a task of finding a number of fingerprint samples from the reference database that match the fingerprint obtained at the unknown location. The WKNN algorithm is a widely used method in fingerprint-based localization applications for low overhead of the system. However, the performance of WKNN closely depends on the choice of the parameter K. A smaller K value can better capture local information, but may result in overfitting. Conversely, a larger K value can better capture global information, but may result in underfitting. For instance, the optimal positioning accuracy is attained with $K = 3$ in [28], while in [43], it is achieved with $K = 4$. As seen in the references [28] and [43], the optimal K will change when the density of the grid locations for obtaining the reference fingerprint database is different. Unfortunately, in real applications, the optimal K may not be always known in advance for the available reference database.

In order to solve this problem, an AWKNN algorithm is proposed to determine the optimal K_{AWE}-Nearest neighbor value in a dynamical way. Unlike the traditional WKNN methods which have the fixed value of K, the AWKNN method would adjust the value of K_{AWE} in a dynamical way in order to achieve the best accuracy at difference places and in different grid networks. Besides, the algorithm does not require any prior knowledge in the offline stage. Therefore, it does not require any additional effort or hardware from the location system. Next, its principle is explained in pseudo code Algorithm 1.

Algorithm 1 Pseudo-Code for Identifying the K Value Dynamically

input : The threshold of two CoADPS r_{th}. The maximum K_{AWE} value K_{max}. The CoADPS between the point to be located and the location points in the fingerprint database.
output: The weight value K_{AWE} of AWKNN algorithm

```
CoADPS ← sort(CoADPS,' descend');
if CoADPS(1) == 1 then K_AWE = 1;
;
else
    for j = 2 : K_max do
        if CoADPS(1)/sum(CoADPS) − CoADPS(j)/sum(CoADPS) ≥ r_th then
            break;
        end
    end
    K_AWE = j − 1;
    return K_AWE;
end
```

In the Algorithm 1, the input parameters r_{th} and K_{max} are obtained by the experiments with the training database in the offline stage. Here r_{th} denotes the threshold of two CoADPS, which is determined by the degree of decline of CoADPS in the training database. K_{max} denotes the maximum value of K_{AWE}. Once the input parameters have been established, the value of K_{AWE} can be determined as follows. First, arrange the input CoADPS in descending order. Next, we can obtain $K_{AWE} = 1$ when there holds $\mathrm{CoADPS}(1) == 1$. $\mathrm{CoADPS}(1) == 1$ means the fingerprints in the database are completely match with the fingerprints at the location to be located. Thus, nearest neighbor position is adopted with $K_{AWE} = 1$. Besides, the determination of the K_{AWE} is based on the fading threshold of CoADPS r_{th}. The coordinates of the points to be located are determined by the reference point corresponding to the CoADPS value of the r_{th} constraint.

Next, with the dynamic K_{AWE} value determined by Algorithm 1, the proposed fingerprint localization by AWKNN method is summarized in Algorithm 2, which is divided into two phases.

1) Offline Mode: First, the target area is segmented into a uniform grid of cells that are equal in size. Wireless sensors or receivers are then used to measure the CFR $\mathbf{H}$ and geographic coordinate information in each cell. Then, the ADFT matrix $\mathbf{D}$ and DDFT matrix $\mathbf{F}$ are constructed based on known system parameters. Next, $\mathbf{H}$, $\mathbf{D}$, and $\mathbf{F}$ are substituted into equation (17) to calculate the

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Algorithm 2 Fingerprint Extraction and Location Estimation

input : The CFR from channel estimation. The geographic coordinates of the reference points.

1 **Offline mode**
2 Grid the geographical area. In addition the number of grid points is N;
3 Obtain the CFR from channel estimation at every grid intersection;
4 Calculate the ADPS on every grid intersection according to (17) ;
5 Store ADPS fingerprint and geographic location in offline database;
6 Obtain the K_{max} and r_{th} with offline database;
7 **Online mode**
8 Select the point for positioning in the geographical area and calculate the ADPS according to the (17). ;
9 Calculate the CoADPS by (26) between the fingerprint and the N offline fingerprints in database;
10 Obtain the K_{AWE} by Algorithm 1, and search through the point pool to find the K_{AWE} nearest reference points;
11 Calculated and return the location estimation $\widehat{loc}$;

output: The estimation location $\widehat{loc}$.

corresponding ADPS for each grid. The ADPS fingerprint and the geographic coordinate information are stored in the offline database. Finally, according to the grid partitioning method and considering a balance between positioning accuracy and algorithm complexity, the K_{max} is determined, usually $K_{max} \in [12, 16]$. And the value of r_{th} is dependent on the extent of CoADPS reduction in offline databases. In general, r_{th} is typically chosen to be less than or equal to $\frac{\text{CoADPS}(1)-\text{CoADPS}(16)}{\text{sum}(\text{CoADPS})}$.

2) Online Mode: First, the wireless sensors or receivers are used to measure the CFR $\mathbf{H}$ in the target area and $\mathbf{H}$ is substitute into (17) to calculate the ADPS. Then, substitute the online collected ADPS and offline stored ADPS into (26), and calculate the CoADPS between them. Finally, obtain the K_{AWE} by Algorithm 1 and calculate the estimation location. The estimated position $\widehat{loc}$ can be calculated as follows:

$$\widehat{loc} = \sum_{i=1}^{K_{AWE}} w_i loc_i \tag{27}$$

where loc_i is the position coordinate of the ith nearest reference point in the training database. And w_i is the weight factors for the ith nearest reference point.

$$w_i = \frac{CoADPS_i}{\sum_{i=1}^{K_{AWE}} CoADPS_i} \tag{28}$$

Remark 2: The proposed ADPS fingerprint accurately extracts TOA, AOA and energy information. Due to spatial consistency as shown in Fig. 3, interpolation or extrapolation methods can be applied to sparse sampling to construct a complete radio map. Additionally, the grid size should be determined based on the fingerprint resolution analysis. The proposed ADPS fingerprint provides the possibility of quick and effortless construction of radio maps.

C. Complexity Analysis and the Summary of Algorithm

Based on the above analysis, the location of online fingerprint is determined. Algorithm 1 summarizes the pseudo-code of K_{AWE} and Algorithm 2 summarizes the proposed AWKNN algorithm. The computational complexity for Algorithm 1 comes from the following parts:

1) The complexity for sorting the CoADPS in step 1 is $\mathcal{O}(N \log N)$;
2) The complexity of the loop in steps 4-9 is $\mathcal{O}(K_{max})$.

The computational complexity for Algorithm 2 mainly comes from step 4,9,10 and 11. It is instructive to note that only one online fingerprint is considered to calculate the algorithm complexity. In step 4, the N ADPS fingerprints are calculated with the complexity of $\mathcal{O}(NN_t^2N_c + NN_cN_t + NN_c^2N_t)$, which is divided into three phase. The complexity for APS is $\mathcal{O}(N_t^2N_c)$, the complexity for DPS is $\mathcal{O}(N_c^2N_t)$ and the the complexity for APS$\odot$DPS is $\mathcal{O}(N_cN_t)$. In step 9, CoADPS is calculated with the complexity of $\mathcal{O}(NN_cN_t)$. The complexity of step 10 is $\mathcal{O}(N \log N)$ as analyzed in Algorithm 1. The complexity of step 11 is $\mathcal{O}(K_{AWE}N_tN_c)$. Accordingly, the total complexity is as follows

$$\mathcal{O}(NN_t^2N_c + NN_c^2N_t + NlogN + NK_{AWE}N_tN_c). \tag{29}$$

Furthermore, we compare the complexity of the proposed ADPS fingerprint localization algorithm with other representative fingerprint positioning methods and pattern matching methods, as illustrated in Table II. Here, L is the the maximum delay compensation. F_l^2 denotes the convolution kernel size, K_{l-1}^{CNN} represents the number of input channels of the $(l-1)$th layer, and K_l^{CNN} signifies the number of filters of the lth layer. p stands for the index of a pool layer, W_p^2 denotes the input size of the pooling layer, K_p^{CNN} refers the number of input channels and P_p^2 indicates the pooling kernel size. Additionally, g represents the index of a fully connected layer, N_{g-1} denotes the number of input neurons, and N_g signifies the number of output neurons.

On one hand, the AWKNN algorithm demonstrates lower complexity and excels in quickly performing matching compared to other classic fingerprint matching algorithms. Pattern matching methods like CNN and Support Vector Machine (SVM) require a substantial amount of time for both model training and result computation. Furthermore, it is possible for the environment to undergo changes, rendering the datasets outdated. On the other hand, the proposed ADPS-AWKNN algorithm introduces minimal complexity while achieving higher accuracy. The CSI-WKNN algorithm exhibits the lowest complexity since it does not involve computations for constructing new fingerprints, but it also has the poorest localization precision.

VI. Performance Evaluation Results

In this section, the performance of the proposed AWKNN localization algorithm based on ADPS fingerprint is evaluated

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TABLE II
THE POSITION COMPUTATIONAL COMPLEX OF DIFFERENT METHOD

Method	Computational complex
ADPS-AWKNN	$\mathcal{O}(NN_t^2N_c + NN_c^2N_t + NlogN + NK_{AWE}N_tN_c)$
ADCPM-WKNN [28]	$\mathcal{O}(NN_t^2N_c + NN_c^2N_t + (2L-1)NN_tN_c + NlogN + NK)$
CSI-WKNN [43]	$\mathcal{O}(NN_tN_c + NlogN + NK)$
ADPS-CNN	$\mathcal{O}(NN_t^2N_c + NN_c^2N_t + \sum_l N_tN_c \cdot F_l^2 \cdot K_l^{\text{CNN}} \cdot K_{l-1}^{\text{CNN}} + \sum_p W_p^2 \cdot K_p^{\text{CNN}} \cdot P_p^2 + \sum_g N_{g-1}^{\text{CNN}} \cdot N_g^{\text{CNN}})$
ADPS-SVM	$\mathcal{O}(NN_t^2N_c + NN_c^2N_t + N^3)$
ADPS-Gaussian process regression	$\mathcal{O}(NN_t^2N_c + NN_c^2N_t + N^3)$

TABLE III
MAJOR WIRELESS PARAMETERS UTILIZED TO GENERATE THE DATASETS

Parameter	Value
Normalized antenna spacing	0.5
Bandwidth	100MHz(outdoor) 100MHz(dynamic) 20MHz(indoor)
Number of path	25
Frequency band	3.5GHz(outdoor) 3.5GHz(dynamic) 2.4GHz(indoor)
Base station antenna	ULA
Antenna elements (N_t)	64
Sub-carrier number (N_c)	64
Number of active BS	1
Grid spacing	1m(outdoor) 0.2m(dynamic) 0.2m(indoor)
Height of BS	6m(outdoor) 6m(dynamic) 2.5m(indoor)
Height of users	2m(outdoor) 1m(dynamic) 1m(indoor)
Number of users to location	5000

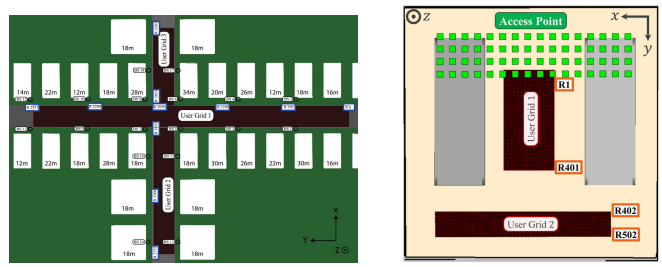
in multiple urban and indoor wireless propagation scenarios. The three public MIMO datasets are adopted in this paper. In addition, the performance is evaluated from multiple perspectives, including localization accuracy, applicable scenarios, the influence of system parameters, time overhead, etc.

The simulation parameters of the transmitted signal are summarized in Table III. The performance of the proposed fingerprint localization method (corresponding to "ADPS-AWKNN") is compared against other fingerprint-based and geometric-based localization methods as follows:

1) CSI-WKNN [43]: The authors directly use the CSI as fingerprint for indoor localization. They use WKNN technology with $K = 4$ to estimate the position in the online fingerprint localization stage.
2) ADCPM-WKNN [28]: The authors propose ADCPM as the fingerprint for localization. The WKNN technology with $K = 3$ is adopted to estimate the position in the online fingerprint localization stage.
3) AOA [46]: The authors use the angle-of-arrivals (AOA) measured at distributed BSs and pseudolinear estimator to triangulate the user position.
4) TOA [47]: The authors utilize time-of-arrivals (TOA) obtained at distributed BSs and output the user position by solving a least-squares fit.
5) AOA-TOA [48]: The authors use hybrid TOA-AOA information and estimator to triangulate the user position.

A. Test Results With Deep MIMO Dataset

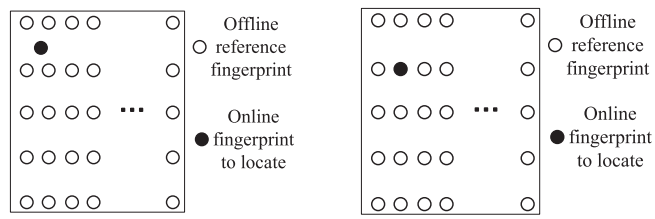
First, the Deep MIMO database [45] is used to verify the universality of the proposed ADPS-AWKNN method.



(a) the outdoor scene, showing the two streets, buildings, with 550m × 36m size of user grid 1.

(b) the indoor scene, showing the two tables, with 4m × 2m size of user grid 1.

Fig. 4. The indoor and outdoor scenario with Deep MIMO database. The ray-tracing scenario is generated using Remcom Wireless InSite [49].



(a) in Deep MIMO database

(b) in ViWi database

Fig. 5. Offline and online fingerprints for Deep MIMO and ViWi database.

The Deep MIMO database is obtained from the ray-tracing simulator, Remcom Wireless InSite [49], which is widely used in mmWave and massive MIMO research at both industry and academia, and has been verified with real-world channel measurements. This database includes indoor scenario with 2.4 GHz frequency OFDM signal, and outdoor urban and dynamic scenario with 3.5 GHz frequency signal. The sizes of the outdoor, dynamic and indoor scenarios for user grid 1 are 550 m × 36 m, 430 m × 15 m and 4 m × 2 m, respectively. Fig. 4 illustrates the structures of the two scenarios.

In the outdoor scenario, the reference fingerprint database is collected from 20387 grid locations, which is formed by 551 equally spaced y-axis direction positions and 37 equally spaced x-axis direction positions. The test database with 5000 test points randomly distribute in the user grid 1 of the outdoor scene. Fig. 5 (a) illustrates the relationship of the test position with the reference grid positions. Besides, Fig. 5 (b) illustrates the relationship of the test position with the reference grid positions with ViWi Dataset [50] which is considered in the next section.

For the Deep MIMO database, since there are four BSs distributed around the localization area, the performance comparisons with all the six algorithms can be performed. Table IV shows the average localization error and root mean square error (RMSE) for the compared algorithms,

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respectively, ADPS-AWKNN, MAP-CSI, ADCPM-WKNN, AOA, TOA, AOA-TOA, and DiSouL by using the Deep MIMO dataset. It can be seen that ADPS-AWKNN achieves the smallest average error and RMSE among compared methods. The smaller average localization error suggests that ADPS-AWKNN has the smallest deviation from the actual positions, while the smaller RMSE indicates that ADPS-AWKNN has a smaller spread of localization errors. Geometric positioning is less accurate than ADPS-AWKNN fingerprint positioning due to the influence of multipath effect. The inaccurate measurement results of TOA in the urban multipath environment lead to the worst positioning effect of TOA. The use of a large number of ULA antennas and the adoption of the AOA positioning method based on multiple incident angle information can improve the accuracy of positioning. The combination of AOA and TOA makes full use of the advantages of these two measuring techniques, further improving the positioning effect. Furthermore, the fingerprint-based method stands out as the sole approach capable of providing reliable localization services using only a single BS.

As previously analyzed, the resolution mainly depends on N_t and N_c. Large values of N_t and N_c result in high resolution. Consequently, the positioning accuracy is improved. Therefore, in Fig. 6, the number of N_t is increased from 4 to 128 in the outdoor scene to evaluate the positioning accuracy by utilizing our proposed localization approach. Taking the 90% point of cumulative distribution function (CDF) as a reference point for reliability, the figure indicates that a single BS with 128 antenna elements can realize location with 1.25 m accuracy. In the future massive MIMO-OFDM systems, the BS will be equipped with more than 100 antenna elements and thus, our proposed method can offer 80% reliability for 0.8 m accuracy and 90% reliability for 1.25 m accuracy. Fig. 7 shows that the positioning accuracy of the proposed localization approach improves with the increasing number of N_c. Taking the 70% point of CDF as a reference point for reliability, the figure indicates that as few as 64 subcarriers can achieve location accuracy of 0.6 m. Afterwards, the positioning accuracy has no significant improvement while $N_t, N_c \geq 64$. Therefore, $N_t, N_c = 64$ is set in the following experiment.

Fig. 8 shows the CDF of the location errors with APS, DPS and ADPS fingerprints by AWKNN approach. The accuracy of ADPS fingerprint is significantly improved over APS and DPS fingerprint. ADPS fingerprint extracts more multipath information (TOA, AOA) from CFR than APS and DPS fingerprints. The accuracy of fingerprint localization increases as more multipath information is decoupled from CFR.

For the dynamic scenario, the main street has 4 lanes of mobile vehicles. These vehicles change their positions with every scene out of 2 captured scenes. As for the propagation model, each channel path can undergo a maximum of 2 reflections and 1 transmission before reaching the receiver. The sizes of the dynamic scenarios for user grid is 430 m × 15 m. The reference fingerprint database is collected from 14632 grid locations, which is formed by 472 equally spaced y-axis direction positions and 31 equally spaced x-axis direction positions.

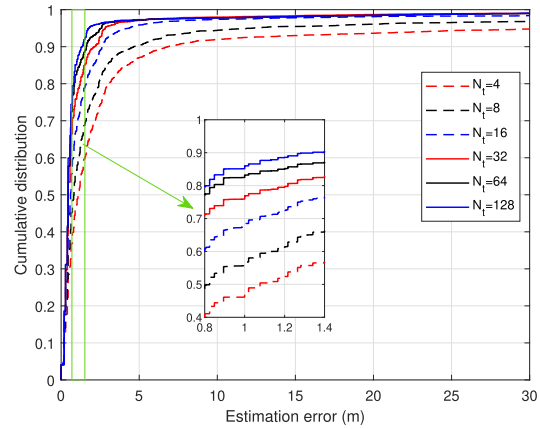

Fig. 6. CDF of the location errors with different N_t by ADPS-AWKNN approach.

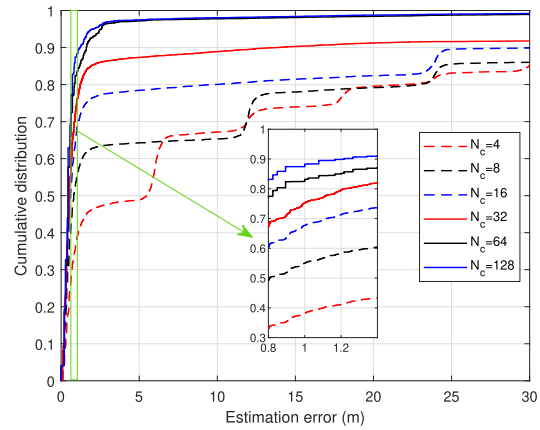

Fig. 7. CDF of the location errors with different N_c by ADPS-AWKNN approach.

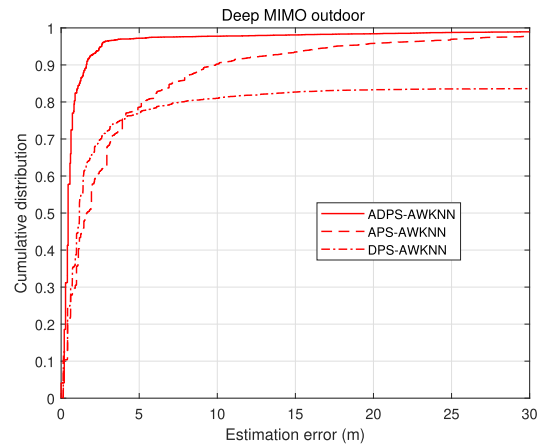

Fig. 8. CDF of the location errors with APS, DPS and ADPS fingerprints by AWKNN approach.

Table V presents the results of the proposed localization compared with ADPS-WKNN and CSI-WKNN in dynamic scenario. The offline fingerprint database originates from the one captured scene, while the online fingerprint database originates from the other captured scene. The proposed ADPS-AWKNN algorithm achieves best accuracy than the others and exhibits a certain level of robustness in various traffic density scenarios (low traffic density for outdoor scenario and high traffic density for dynamic scenario). The offline and online

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TABLE IV
THE INFLUENCE OF DIFFERENT LOCALIZATION METHODS ON POSITIONING RESULTS

	Fingerprint Localization Method			Geometric Measurement Localization Method		
	ADPS-AWKNN	ADCPM-WKNN	CSI-WKNN	AOA	TOA	AOA-TOA
Average error (m)	2.662	111.841	146.167	6.113	115.683	5.956
RMSE	18.507	170.621	180.751	46.252	231.000	46.250

TABLE V
THE INFLUENCE OF DIFFERENT LOCALIZATION METHODS ON POSITIONING RESULTS IN DYNAMIC SCENARIO

	ADPS-AWKNN	ADCPM-WKNN	CSI-WKNN
Average error (m)	2.152	60.816	109.033
RMSE	13.549	93.371	124.888

fingerprint databases collected from different capture scenarios with the same traffic density can still be well matched and located using the proposed algorithm. When faced with locating scenes of different traffic densities, the ADPS-AWKNN fingerprint localization method requires selection of the corresponding traffic density offline fingerprint database for accurate positioning.

For the indoor scenario, the reflection paths from the floor and the ceiling are indeed considered. Each channel path can undergo a maximum of two reflections before reaching the receiver. Only static objects, such as walls, tables, and glass windows, are considered in this scenario. The dimensions of the indoor room are 10 m in length, 10 m in width, and 5 m in height. The table is 6 m long and 2 m wide. The glass window is 3 m wide and 1.7 m tall. In the indoor scenario, the reference fingerprint database is formed by 20301 grid samples with 4 m $\times$ 2 m size, which are constituted by 201 equally spaced y-axis direction positions and 101 equally spaced x-axis direction positions. The test database with 5000 test points randomly distribute in the user grid 1 of the indoor scene. Fig. 5(a) illustrates the relationship of the test position with the reference grid positions.

Fig. 9 shows the performance comparison among the proposed ADPS-AWKNN method, the ADCPM-WKNN method and the CSI-WKNN method. It can be seen that the localization accuracy for the 90% reliability for these three methods are 0.02 m, 0.82 m, and 2.49 m, respectively. Clearly, the proposed ADCPM-WKNN achieves best accuracy than the others.

With the above Deep MIMO datdabase setting, our proposed method is applicable to not only WiFi and 5G signals, but also indoor and outdoor scenes. The localization algorithm is robust to the environment and signal type (or the frequency band of the signal). In order to better compare our proposed method with other type of localization methods, the ViWi Dataset which involves more complex positioning scenes than Deep MIMO database is adopted in the next experiments.

B. Test Results With ViWi Database

Another public mmWave MIMO dataset, called ViWi Dataset [50], is also utilized in this study to test the localization performance of the proposed algorithm. This dataset utilizes 3D modeling and ray-tracing simulators that generate urban wireless data for a carrier frequency of 60 GHz with a 0.2

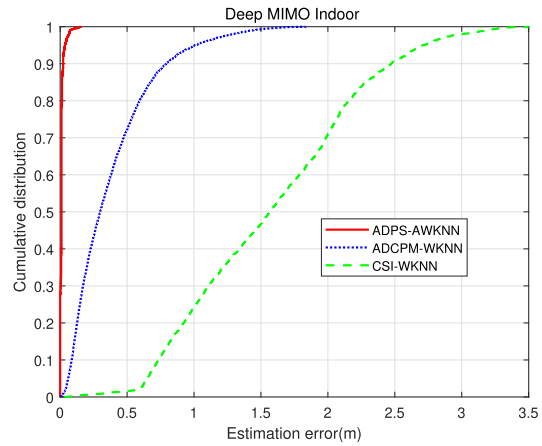
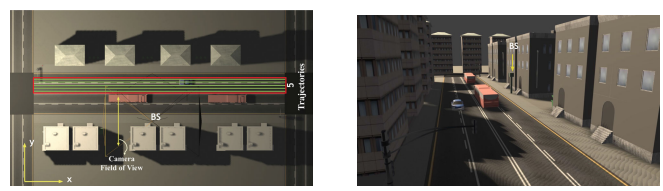


Fig. 9. The CDF for the location errors with different methods in indoor scene.



(a) the top view of mixed environment (b) the perspective view of mixed environment

Fig. 10. The top and perspective views of the ViWi dataset scenario for LOS environment [50]. The top view (on the left) shows the 5 trajectories of the car, the stationary blockages (buses) and the location of the BS. The perspective view (on the right) shows an overview of the street, building, car, buses, and the BS.

GHz signal bandwidth. Other wireless parameters employed to generate the dataset are listed in Tale III. Fig. 10 shows the overview of the buildings, roads, buses, and the BS in the mixed environment with LOS and NLOS paths with a size of 90 m $\times$ 4 m. Fig. 11 shows the overview of the buildings, roads, and the BS in the LOS environment with a size of 90 m $\times$ 4 m. The height of the BS of the ViWi database is 5 m. The height of the users is 1.5 m. In this dataset, there are a total of 5000 samples at evenly spaced grid locations. These locations are constructed by 5 equally spaced horizontal positions and 1000 equally spaced vertical positions. The spacing for adjacent horizontal positions and vertical positions are 1 m and 0.09 m, respectively.

Because the locations of the samples in this dataset cannot be randomly changed, the leave-one-out cross validation method [51] is adopted in the localization accuracy evaluation. This method evaluates the accuracy of the tested localization algorithms by removing a sample from the dataset and then tries to estimate its location by using the remaining samples as the reference dataset. This process is repeated for all the samples and the final positioning results are the set of estimates for all sample positions. And the distribution of offline and online fingerprints are shown in Fig. 5 (b). It can be inferred

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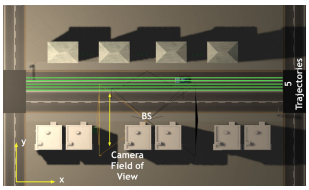
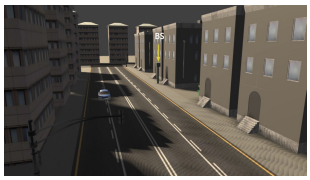


(a) the top view of LOS environment



(b) the perspective view of LOS environment

Fig. 11. The top and perspective views of the ViWi dataset scenario for LOS environment [50]. The top view (on the left) shows the 5 trajectories of the car and the location of the BS. The perspective view (on the right) shows an overview of the street, building, car, and the BS.

that this evaluation approach is more conservative than that used in the test with Deep MIMO database of Fig. 5 (a), since there is no corresponding reference sample in the training dataset for the estimated position.

As to the ViWi dataset, the geometric-based algorithms of AOA, TOA, and AOA-TOA are not included because there is only one BS in this scenario. However, we have added another algorithm called MAP-CSI [31]. This algorithm extracts the AOA and TOA from the angle delay profile matrix, and it helps localize the users with the assistance of the environment map. Therefore, MAP-CSI can be taken as the method of jointly utilizing channel fingerprints and geometric measurements. It needs to point out that the MAP-CSI method involves k-means algorithm in which the initial cluster center is randomly selected, thus resulting in random position results. For this reason, we adopt 1000 Monte Carlo runs to obtain its average localization accuracy in the simulation test.

Fig. 12 shows the CDF of the position estimation error for the compared algorithms, respectively, ADPS-AWKNN, ADCPM-WKNN, CSI-WKNN and MAP-CSI, by using the ViWi dataset of the mixed environment. Taking the 70% point of the CDF as the localization accuracy criteria, it can be seen that ADPS-AWKNN, ADPCM-WKNN, CSI-WKNN and MAP-CSI achieve the accuracy of 0.179 m, 0.735 m, 41.758 m and 9.098 m, respectively. The CSI-WKNN method provides the worst accuracy for multiple-path information is deeply coupled in CSI. The MAP-CSI method requires both CFR data and environment map information, but it still suffers from several significant estimation errors due to the absence of a direct LOS path. The ADCPM and ADPS fingerprint location algorithm can provide better location accuracy than map assistant location algorithm due to their capability to capture subtle signal variations and compensate for NLOS paths and multipath effects. Besides, the proposed ADPS-AWKNN methods can achieve more accurate positioning accuracy than ADCPM-WKNN algorithm for the high resolution fingerprint and AWKNN localization method.

Fig. 13 shows the CDF of the position estimation error for the compared algorithms: ADPS-AWKNN, ADCPM-WKNN, CSI-WKNN, and MAP-CSI. The evaluation was performed using the ViWi dataset in a LOS environment. Taking the 70% point of the CDF as the localization accuracy criteria, we can observe that ADPS-AWKNN, ADPCM-WKNN, CSI-WKNN, and MAP-CSI achieve the accuracy of 0.221 m, 0.608 m, 47.030 m and 5.867 m, respectively. The proposed fingerprint localization method achieves lower positioning

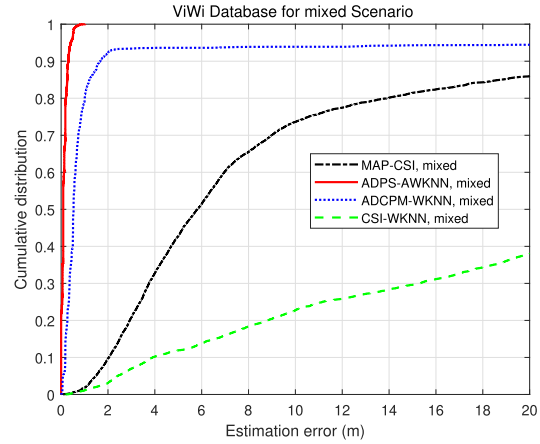


Fig. 12. CDF of the localization errors with different methods in ViWi Database for mixed scene.

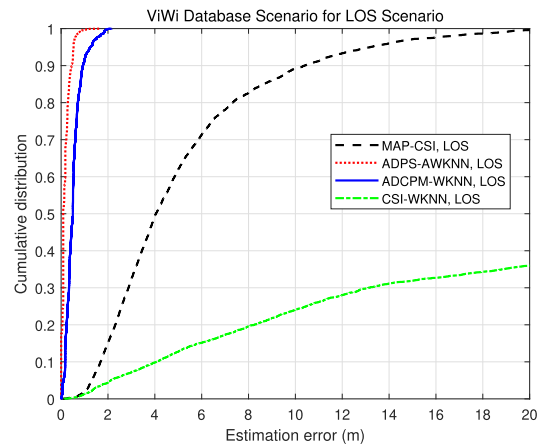


Fig. 13. CDF of the localization errors with different methods in ViWi Database for LOS scene.

accuracy in comparison to the results obtained in the mixed environment. This is mainly due to that the algorithm is designed mainly for rich scattering environments. The mixed environment has more rich scatters than LOS environment. Moreover, the performance of CSI fingerprints and ADCPM fingerprints is enhanced in LOS environments. CSI fingerprints and ADCPM fingerprints may not effectively extract multipath information in the channel state information, and their localization performance heavily depends on environmental factors for direct path. In LOS environments, with fewer or no obstacles, the signal propagation paths are relatively direct and stable, making it easier to capture and extract signal features. Furthermore, the MAP-CSI method tends to yield larger errors in NLOS samples due to the obstruction caused by blockers, which can reduce the number of points in the cluster nearest to the user by obstructing some of the critical paths.

Taking the 70% point of CDF as a reference point for reliability, Fig. 14 shows the localization accuracies of the ADPS-AWKNN, ADPCM-WKNN, CSI-WKNN and MAP-CSI with different numbers of array antenna elements and different numbers of subcarriers. Compared with the MAP-CSI, CSI-WKNN and ADPCM-WKNN method, the positioning error of ADPS-AWKNN method has decreased by at least 7 m, 31 m and 0.3 m for all values of N_t and N_c. Furthermore, we observe that increasing N_t and N_c in

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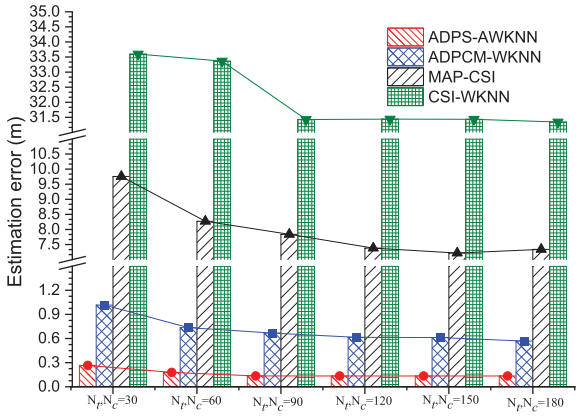

Fig. 14. CDF of the localization errors with different methods in ViWi Database scene of mixed environment.

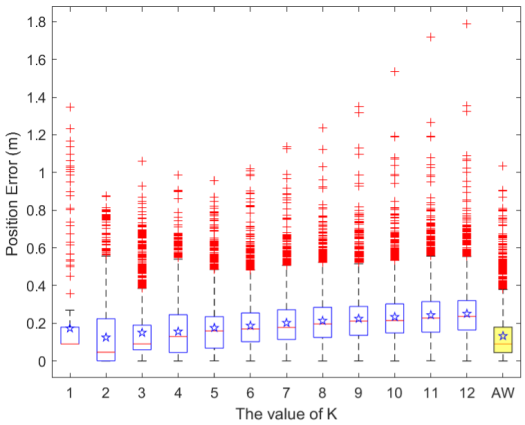

Fig. 15. The distribution of positioning error of AWKNN and WKNN algorithm with different K value.

ADPS-AWKNN approach reduces the position error, which is consistent with the resolution analysis in section IV.

Fig. 15 shows the distribution of positioning error of the proposed AWKNN algorithm in comparison with the fixed K value WKNN method when both using the ADSP fingerprints. The notation "AW" in Fig. 15 stands for the AWKNN method. The estimation errors are displayed by box charts. Each box contains seven metrics, i.e., the outlier (the red plus sign), the upper tentacle (1.5 box), the upper quartile (75%), the average (small five-pointed star),the median,the next quartile (25%), and the next tentacles (−1.5 box). As shown in Fig. 15, the minimal average position error occurs when K equals to 2 for the traditional WKNN method. If choosing an improper K value when using this method, the average position error will increase inevitably. Unfortunately, the optimal K value for different application scenarios is usually different and normally cannot be known in advance. However, the proposed AWKNN properly solves this dilemma and achieves better performance than WKNN method with $K = 2$.

The position accuracy is different in the Deep MIMO database and the ViWi database when the same position methods is utilized. Because the size of gride also decide the position accuracy. Under the premise of meeting the fingerprint resolution (21) and (25), the smaller the grid partition, the higher the positioning accuracy will be.

Positioning performance depends not only on positioning accuracy but also on position time overhead. For the above methods, the fingerprints and the corresponding coordinates are calculated and save using MATLAB R2023a. Our simulation is carried out on a workstation equipped with Intel(R) Core(TM) i5-10400F CPU and 16 GB RAM. During the simulation, the unknown position of 5000 points was estimated, and the average positioning time of each unknow position point was calculated. The position time overhead of different position approach is shown in Table VI. As we can see, the fingerprint localization method is able to make a location estimation in less than 1s and outperforms the others. While ADPS-AWKNN enhances the positioning accuracy, it incurs an additional time load compared to ADPCM-WKNN, since it requires time to find the optimal K value. Both the fingerprint localization method and map assistant location method suffer a time complexity increase as the BS antennas and sub-carriers increase.

TABLE VI
THE POSITION TIME OVERHEAD OF FOUR POSITION METHOD, UNIT: S

method	N_t, N_c					
	30	60	90	120	150	180
ADPS-AWKNN	0.00615	0.0829	0.196	0.281	0.684	0.922
ADCPM-WKNN	0.00655	0.0318	0.0631	0.170	0.180	0.426
CSI-WKNN	0.110	0.445	0.932	1.1497	1.303	1.731
MAP-CSI	6.958	8.177	9.522	12.001	15.278	19.475

C. Test Results With MaMIMO CSI Dataset

Real-test experiment public MIMO dataset [52], called MaMIMO CSI, is also utilized to test the localization performance of the proposed algorithm. The experiment system consists of a BS that is equipped with 64 ULA antennas, and four user equipments (UE) each of which has a single dipole antenna. During the measurements, the BS was configured to use a center frequency of 2.61 GHz with a bandwidth of 20 MHz. The BS ULA is at a height of 1 m above the floor, and the height of UE is 40 cm above the floor. The size of the indoor LOS scenarios includes four user grids and 1.25 m by 1.25 m area for each user grid. Using this set-up, the area is scanned with 25 mm intervals, resulting in a dataset containing 10000 CSI samples. The test database consists of 5000 test points randomly distributed among the four user grids. The relationship of the test position with the reference grid positions is illustrated in Fig. 5 (a).

For the MaMIMO CSI dataset, we only compare ADPS-AWKNN with ADPS-2D-CNN [29], ADPS-3D-CNN [36], ADCPM-WKNN, and CSI-WKNN, because this dataset does not provide any AOA or TOA measurements. Fig. 16 depicts the CDF of the position estimation error for the compared algorithms, respectively, ADPS-AWKNN, ADPS-2D-CNN, ADPS-3D-CNN, ADCPM-WKNN, and CSI-WKNN. These methods achieve the average accuracy of 9.240 mm, 129.579 mm, 187.723 mm, 680.160 mm, and 1053.230 mm, respectively. The ADPS-AWKNN algorithm achieves the highest localization accuracy due to its more precise extraction of multipath channel information, accurate fingerprint similarity matching criterion, and adaptive WKNN K-value selection algorithm. The AWKNN algorithm achieves better localization accuracy than the CNN algorithm due to

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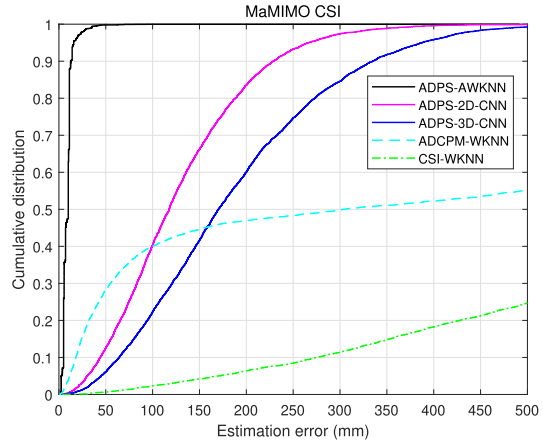

Fig. 16. The impact of various localization methods on positioning results in the MaMIMO CSI dataset.

its suitability for ADPS fingerprints and robustness in various environments. ADPS fingerprints are two-dimensional and exhibit continuity in changing geographic locations, allowing the WKNN algorithm to utilize similar features from an offline database to determine the fingerprint's position. Additionally, 2D-CNN fully utilizes the information in the two-dimensional fingerprint, whereas 3D-CNN, designed for three-dimensional data, faces challenges in capturing relevant information from the two-dimensional ADPS fingerprint due to data dimension mismatch. As a result, the localization performance of the 2D-CNN network outperforms that of the 3D-CNN network. In addition, neural networks must design and adjust network parameters for different training fingerprints and datasets, requiring abundant computing resources, which limits practical applications. The performances of ADCPM and CSI degrade significantly in practical scenario databases. The ADPS-AWKNN algorithm demonstrates robustness in both practical and simulated scenario datasets.

VII. Conclusion

In this study, a localization method by using the fingerprints of MIMO OFDM signals was presented. A new type of multipath channel fingerprint, named as ADPS, was proposed. Then, the AWKNN algorithm was proposed to perform position estimation by utilizing the ADPS fingerprints. Evaluations with Deep MIMO dataset, ViWi dataset and MaMIMO CSI dataset were performed to validate the performance superiority of the proposed method to the current algorithms reported in references. The proposed ADPS fingerprints have high angle resolution and delay resolution. On the one hand, compared to the traditional WKNN algorithm, the AWKNN algorithm offers a more flexible selection of K values, resulting in more accurate positioning. On the other hand, compared to machine learning algorithms, the AWKNN algorithm has a lower algorithmic complexity. Besides, the proposed fingerprint localization method is suitable for both indoor and outdoor scenarios and does not require a large number of receiving antennas.

It is expected that more complex pattern recognition algorithms based on deep neural networks can be utilized with ADPS fingerprint to further improve the positioning accuracy in the future. Although the proposed algorithm is robust to dynamic scenarios, in future work, we want to further design a general database to cover different traffic densities scenarios, while we design offline database selection algorithms to adaptively select appropriate offline fingerprint databases for localization based on the traffic densities of online fingerprints. Furthermore, the quick and effortless method for radio map construction, based on the proposed ADPS fingerprint, is a highly meaningful research direction for the future.

Appendix A

According to (8), The element $\sum_{r=1}^{N_t} d_{m,r}h_{r,l}$ in the APS matrix can be redefined as follows

$$\begin{aligned}&\sum_{r=1}^{N_t} d_{m,r}h_{r,l}\\&=\sum_{p=1}^{P} a_p\frac{1}{\sqrt{N_t}}e^{-j2\pi\frac{ln_p}{N_c}}\sum_{r=0}^{N_t-1} e^{-j2\pi r\frac{d_s(\cos\phi_p+\cos\pi\frac{\mathrm{m}}{N_\mathrm{t}})}{\lambda_c}}\\&=\sum_{p=1}^{P} \alpha_p\frac{1}{\sqrt{N_t}}e^{-j2\pi\frac{ln_p}{N_c}}I_m,\end{aligned}\tag{A.1}$$

where

$$I_m=\sum_{r=0}^{N_t-1} e^{-j2\pi r\frac{d_s(\cos\phi_p+\cos\pi\frac{\mathrm{m}}{N_\mathrm{t}})}{\lambda_c}}.\tag{A.2}$$

Here $m, r = 0, \ldots, N_t-1, l = 0, \ldots, N_c-1$.

Therefore, $|I_m|$ can be expressed as

$$\begin{aligned}|I_m|=|1+e^{-j2\pi\frac{d_s(\cos\phi_p+\cos\pi\frac{\mathrm{m}}{N_\mathrm{t}})}{\lambda_c}}+e^{-j4\pi\frac{d_s(\cos\phi_p+\cos\pi\frac{\mathrm{m}}{N_\mathrm{t}})}{\lambda_c}}\\+\ldots+e^{-j2\pi(N_t-1)\frac{d_s(\cos\phi_p+\cos\pi\frac{\mathrm{m}}{N_\mathrm{t}})}{\lambda_c}}|.\end{aligned}\tag{A.3}$$

Considering that each term in (A.3) exits unit norm, thus there holds

$$|I_m|=\left|\sum_{r=0}^{N_t-1} e^{-j2\pi r\frac{d_s(\cos\phi_p+\cos\pi\frac{\mathrm{m}}{N_\mathrm{t}})}{\lambda_c}}\right|\le N_t.\tag{A.4}$$

For every $0\le r\le N_t-1$ in (A.1), if and only if $e^{-j2\pi r\frac{d_s(\cos\phi_p+\cos\pi\frac{\mathrm{m}}{N_\mathrm{t}})}{\lambda_c}}=1$, the magnitude of I_m attains its maximal value N_t. While $\frac{d_s(\cos\phi_p+\cos\pi\frac{\mathrm{m}}{N_\mathrm{t}})}{\lambda_c}\in Z$, $\sum_{r=1}^{N_c} d_{m,r}h_{r,l}$ will reach the maximum value $\sum_{p=1}^{P}\alpha_p\sqrt{N_t}e^{-j2\pi\frac{ln_p}{N_c}}$ and we can attain $\cos\phi_p+\cos\pi\frac{\mathrm{m}}{N_\mathrm{t}}=\frac{\lambda_c}{d_s}n, n\in Z$. To avoid antenna coupling, $\frac{d_s}{\lambda_c}\le\frac{1}{2}$ is assumed. Besides, $\cos\phi_p$ and $\cos\pi\frac{\mathrm{m}}{N_\mathrm{t}}$ satisfy the condition that $\cos\phi_p, \cos\pi\frac{\mathrm{m}}{N_\mathrm{t}}\in(-1,1)$. For these conditions, we can obtain that $\frac{\lambda_c}{d_s}n=0$. By letting

$$m_{\lim}=\underset{0\le m\le N_t-1}{\arg\max}\ |I_m|,\tag{A.5}$$

we have

$$m_{\lim}=\lfloor N_t-N_t\frac{\phi_p}{\pi}\rceil.\tag{A.6}$$

If $\frac{N_t d_s\cos\phi_p}{\lambda_c}\notin Z$, the maximum value cannot be reached, but $|I_m|$ is still close to N_t since each $e^{-j2\pi r\frac{d_s(\cos\phi_p+\cos\pi\frac{\mathrm{m}}{N_\mathrm{t}})}{\lambda_c}}$

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in (A.1) is close to 1. Meanwhile, while N_t tends to be infinity, the peak of $|I_{m_{lim}}|$ will account for most of the signal power and the power that leakages to other $|I_m|$ is negligible [53].

Thus, we can simply (A.1) as follows

$$\sum_{r=1}^{N_c} d_{m,r} h_{r,l} = \sum_{p=1}^{P} \alpha_p \sqrt{N_t} e^{-j2\pi \frac{ln_p}{N_c}} \delta(m - (N_{\mathrm{t}} - \frac{N_t \phi_p}{\pi})). \tag{A.7}$$

This concludes the derivation process.

Appendix B

First the properties of $|\mathrm{APS_R(m)}|^2$ is analyzed. $|\mathrm{APS_R(m)}|^2$ has two main terms $W_{N_t}^2\left(\frac{\pi d_s(\cos\phi_{p_i}+\cos\frac{\pi m}{N_t})}{\lambda_c}\right)$ and $W_{N_t}^2\left(\frac{\pi d_s(\cos\phi_{p_j}+\cos\frac{\pi m}{N_t})}{\lambda_c}\right)$ which both have the max value N_t^2 at $cos\frac{\pi m_{maxi}}{N_t} = -cos\phi_{p_i}$ and $cos\frac{\pi m_{maxj}}{N_t} = -cos\phi_{p_j}$, respectively. The overlapping point of the main terms $W_{N_t}^2\left(\frac{\pi d_s(\cos\phi_{p_i}+\cos\frac{\pi m}{N_t})}{\lambda_c}\right)$ and $W_{N_t}^2\left(\frac{\pi d_s(\cos\phi_{p_j}+\cos\frac{\pi m}{N_t})}{\lambda_c}\right)$ satisfies $cos\pi\frac{m_{cross}}{N_t} = -\frac{cos\phi_{p_i}+cos\phi_{p_j}}{2}$.

Then according to the properties of $|\mathrm{APS_R(m)}|^2$, the angular resolution is derived as follows. The arrival angle information of two different paths is distinguishable when $|\mathrm{APS_R(m)}|^2$ has two peaks.

On the one hand, the main lobes of the two main terms are not overlapping. Thus, it is derived that $\left|\cos\phi_{p_i} - \cos\phi_{p_j}\right| > \frac{2}{N_t d_s/\lambda_c}$.

On the other hand, when the main lobes of the two main terms are overlapping ($\left|\cos\phi_{p_i} - \cos\phi_{p_j}\right| \leq \frac{2}{N_t d_s/\lambda_c}$), the value of $|\mathrm{APS_R(m)}|^2$ at the overlapping point m_{cross} is less than N_t^2. Substitute $cos\pi\frac{m_{cross}}{N_t} = -\frac{cos\phi_{p_i}+cos\phi_{p_j}}{2}$ into (20), we obtain as follows

$$\begin{aligned}
&|\mathrm{APS_R(m_{cross})}|^2 \\
&= \left(W\left(\pi\frac{d_s(\frac{\cos\phi_{p_i}-\cos\phi_{p_j}}{2})}{\lambda_c}\right)\right. \\
&\quad \left. -W\left(\pi\frac{d_s(\frac{\cos\phi_{p_i}-\cos\phi_{p_j}}{2})}{\lambda_c}\right)\right)^2 \\
&\quad + \left(2 + 2\cos\left(\pi(N_t-1)\frac{d_s(\cos\phi_{p_i} - \cos\phi_{p_j})}{\lambda_c}\right)\right) \\
&\quad \times W\left(\pi\frac{d_s(\frac{\cos\phi_{p_i}-\cos\phi_{p_j}}{2})}{\lambda_c}\right)^2.
\end{aligned} \tag{B.1}$$

In order to make the value of $|\mathrm{APS_R(m)}|^2$ always be less than N_t^2, then there holds $\left(2 + 2\cos\left(\pi(N_t-1)\frac{d_s(\cos\phi_{p_i}-\cos\phi_{p_j})}{\lambda_c}\right)\right) \leq 1$. Thus, it can be derived that $\frac{2}{3}\frac{\lambda_c}{d_s(N_t-1)} < |\cos\phi_{p_i} - \cos\phi_{p_j}| < \frac{4}{3}\frac{\lambda_c}{d_s(N_t-1)}$. Considering that there also need satisfy $|\cos\phi_{p_i} - \cos\phi_{p_j}| \leq 2$. While $N_t \gg 1$, there holds $2 - \frac{4}{3}\frac{\lambda_c}{d_s(N_t-1)} = \frac{2(N_t-3)}{3N_t(N_t-1)d_s/\lambda_c} \geq 0$. Therefore, when the main lobes of the two main terms are overlapping, the final angle resolution is $\left(2 + 2\cos\left(\pi(N_t-1)\frac{d_s(\cos\phi_{p_i}-\cos\phi_{p_j})}{\lambda_c}\right)\right) \leq 1$.

This concludes the derivation process.

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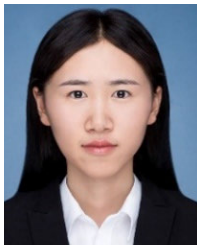

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